
Construction Specification 8 — Mobilization and Demobilization

Title 210 – Engineering

Part 642 – Specifications for Construction Contracts

Subpart B – National Construction Specifications

June 2024

A. Scope

1. The work consists of the mobilization and demobilization of the contractor's forces and equipment necessary for performing the work required under the contract. It does not include mobilization and demobilization for specific items of work for which payment is provided elsewhere in the contract. Mobilization will not be considered as work in fulfilling the contract requirements for commencement of work.

B. Equipment and Material

1. Mobilization includes all activities and associated costs for transportation of contractor's personnel, equipment, and operating supplies to the site; establishment of offices, buildings, and other necessary general facilities for the contractor's operations at the site; premiums paid for performance and payment bonds including coinsurance and reinsurance agreements as applicable; and other items specified in Section D of this specification.
2. Demobilization includes all activities and costs for transportation of personnel, equipment, and supplies not required or included in the contract from the site, including the disassembly, removal, and site cleanup of offices, buildings, and other facilities assembled on the site specifically for this contract.
3. This work includes mobilization and demobilization required by the contract at the time of award. If additional mobilization and demobilization activities and costs are required during the performance of the contract as a result of changed, deleted, or added items of work for which the contractor is entitled to an adjustment in contract price, compensation for such costs will be included in the price adjustment for the item or items of work changed or added.

C. Payment

1. Payment will be made as the work proceeds after presentation of paid invoices or documentation of direct costs by the contractor showing specific mobilization and demobilization costs and supporting evidence of the charges of suppliers, subcontractors, and others. When the total of such payments is less than the lump-sum contract price, the balance remaining will be included in the final contract payment. Payment of the lump sum contract price for mobilization and demobilization will constitute full compensation for completion of the work.

2. Payment will not be made under this item for the purchase costs of materials having a residual value, the purchase costs of materials to be incorporated in the project, or the purchase costs of operating supplies.

D. Items of Work and Construction Details

3. Bid Item - #1 – Mobilization/Demobilization
4. This Item shall consist of the following
 - a. Mobilization and demobilization of the contractor's work force and equipment.
 - b. Purchase of bonds if required by the terms of the contract.
 - c. Purchase of liability insurance coverage for accidental injury to personnel on the work site and accidental damage to property located within the work sites and damaged during the performance period of the contract.
 - d. Costs of moving equipment to the work sites and removal of equipment after completion of the work.
 - e. The contractor shall assume responsibility for protection of existing roads, overhead and/or buried utilities and fences that may be affected by the work. Prior to start of construction, contractor must provide COTR with JULIE – one call dig ticket number
 - f. The contractor is responsible for grading access. All access routes shall be returned to original condition after work is complete.
 - g. Contractor shall provide COTR notification of minimum of 5 business day prior to commencement of construction. Contact COTR at 217-208-4020
 - h. All invoices submitted by the contractor shall be subject to approval of the engineer.
 - i. Measurement and Payment: Payment will be made by the method in Section C. Payment

Construction Specification 6 — Seeding, Sprigging, and Mulching

Title 210 - Engineering

Part 642 – Specifications for Construction Contracts

Subpart B – National Construction Specifications

June 2024

A. Scope

1. The work consists of preparing the area for treatment; furnishing and placing seed, sprigs, mulch, fertilizer, inoculant, lime, and other soil amendments; and anchoring mulch in designated areas as specified.
2. The following BioPreferred® product categories are applicable to this specification:
 - a. Mulch and compost materials
 - b. Erosion control materials
 - c. Fertilizers
 - d. Agricultural spray adjuvants

B. Material

1. Seed.
 - a. All seed must conform to the current rules and regulations of the state where it is being used and must be from the latest crop available. It must meet or exceed the standard for purity and germination listed in Section G.
 - b. Seed must be labeled in accordance with the state laws and the U.S. Department of Agriculture rules and regulations under the Federal Seed Act in effect on the date of invitations for bids. Bag tag figures are evidence of purity and germination. No seed will be accepted with a test date of more than 9 months before the delivery date to the site.
 - c. Seed that has become wet, moldy, or otherwise damaged in transit or storage will not be accepted. The percent of noxious weed seed allowable must be as defined in the current state laws relating to agricultural seeds. Each type of seed must be delivered in separate sealed containers and fully tagged unless an exception is granted in writing by the contracting officer.
2. Fertilizer. Unless otherwise specified, the fertilizer must be a commercial-grade fertilizer. It must meet the standard for grade and quality specified by state law. Where fertilizer is furnished from bulk storage, the contractor must furnish a supplier's certification of analysis and weight. When required by the contract, a representative sample of the fertilizer must be furnished to the contracting officer for chemical analysis.

3. Inoculants. The inoculant for treating legume seeds must be a pure culture of nitrogen-fixing bacteria prepared specifically for the species and must not be used later than the date indicated on the container or as otherwise specified. A mixing medium, as recommended by the manufacturer, must be used to bond the inoculant to the seed. Two times the amount of the inoculant recommended by the manufacturer must be used (unless seed is applied using a hydraulic seeder; then four times the amount must be used). Seed must be sown within 24 hours of treatment and must not remain in the hydraulic seeder longer than 4 hours.
4. Lime and Other Soil Amendments. Lime must consist of standard ground agriculture limestone or approved equivalent. Standard ground agriculture limestone is defined as ground limestone meeting current requirements of the state's Department of Agriculture. Other soil amendments must meet quality criteria and application requirements specified in Section G.
5. Mulch Tackifiers. Asphalt emulsion tackifiers must conform to the requirements of ASTM D977, Specification for Emulsified Asphalt. The emulsified asphalt may be rapid setting, medium setting, or slow setting. Non-asphaltic tackifiers required because of environmental considerations must be as specified in Section G.
6. Straw Mulch Material. Straw mulch must consist of wheat, barley, oat or rye straw, hay, grass cut from native grasses, or other plants as specified in Section G. The mulch material must be air-dried reasonably light in color and must not be musty, moldy, caked, or otherwise of low quality. The use of mulch that contains noxious weeds is not permitted. The contractor must provide a method satisfactory to the contracting officer for determining weight of mulch furnished.
7. Other Mulch Materials. Mulching materials, such as wood cellulose fiber mulch, mulch tackifiers, synthetic fiber mulch, netting, and mesh may be required for specialized locations and conditions. These materials, when specified, must be accompanied by the manufacturer's recommendations for methods of application.

C. Seeding Mixtures, Sod, Sprigs, and Dates of Planting

1. The application rate per acre for seed mixtures, sprigs, or sod and the date of seeding or planting must be as shown on the plans or as specified in Section G.

D. Seedbed Preparation and Treatment

1. Dress areas to be treated to a smooth, firm surface. On sites where equipment can operate on slopes safely, adequately loosen and smooth the seedbed (4 to 6 inches deep). Depending on soil and moisture conditions, disking, cultipacking, or both, may be necessary to properly prepare a seedbed. Where equipment cannot operate safely, prepare the seedbed by methods like scarifying, providing a roughened soil surface by hand so that broadcast seed will remain in place.
2. If seeding is to be accomplished immediately following construction operations, seedbed preparation may not be required except on a compacted, polished, or freshly cut soil surface.

3. Rocks larger than 6 inches in diameter, trash, weeds, and other debris that will interfere with seeding or maintenance operations must be removed or disposed of as specified in Section G.
4. Discontinue seedbed preparation when soil moisture conditions are not suitable for the preparation of a satisfactory seedbed as determined by the responsible engineer.

E. Seeding, Sprigging, Fertilizing, Mulching, and Stabilizing

1. Perform all seeding or sprigging operations in such a manner that the seed or sprigs are applied in the specified quantities uniformly in the designated areas. The method and rate of seed application must be as specified in Section G. Unless otherwise specified, seeding or sprigging must be accomplished within 2 days after final grading is completed and approved.
2. Apply fertilizer, lime, and other soil amendments as specified in Section G. When specified, the fertilizer and soil amendments must be thoroughly incorporated into the soil immediately following surface application.
3. The rate, amount, and kind of mulching or mesh must be as specified in Section G. Apply mulches uniformly to the designated areas. They must be applied to areas seeded not later than 2 working days after seeding has been performed. Stabilize straw mulch material within 24 hours of application using a mulch crimper or equivalent anchoring tool or by a suitable tackifier. When the mulch crimper or equivalent anchoring tool is used, it must have straight blades and be the type manufactured expressly for and capable of firmly punching the mulch into the soil. Where the equipment can be safely operated, it must be operated on the contour. Use hand methods where equipment cannot safely operate to perform the work required.
4. Apply the tackifier uniformly over the mulch material at the specified rate or inject it into the mulch material as it is being applied. Stabilizing materials like mesh or netting must be applied smoothly but loosely on the designated areas. The edges of these materials must be buried or securely anchored using spikes or staples as specified in Section G.
5. The contractor must maintain the mesh or netting areas until all work under the contract has been completed and accepted. Maintenance must consist of the repair of areas damaged by water erosion, wind, fire, or other causes. Such areas must be repaired to reestablish the intended condition and to the design lines and grades required by the contract. The areas must be re-fertilized, reseeded, and re-mulched before the new application of the mesh or netting.

F. Measurement and Payment

1. Method 1. For items of work for which specific unit prices are established in the contract, each area treated is measured as specified in Section G and area is calculated to the nearest 0.1 acre. Payment for treatment is made at the contract unit price for the designated treatment, which will constitute full compensation for completion of the work.

- a. When specified as an item of work, mesh or netting is measured to the nearest square yard of surface area covered and accepted. Payment is made at the contract unit price and will constitute full compensation for completion of the work.
2. Method 2. For items of work for which specific lump-sum prices are established in the contract, the quantity of work will not be measured for payment. Payment for this item is made at the contract lump-sum price for the item and will constitute full compensation for the completion of the work.
3. Method 3. For items of work for which lump-sum prices are established in the contract, payment is made as the work proceeds. Progress payments will be determined as specified in Section G. Payment of the lump-sum contract price will constitute full compensation for completion of the work.
4. All Methods. The following provisions apply to all methods of measurement and payment. Compensation for any item of work described in the contract but not listed in the bid schedule is included in the payment for the item of work to which it is made subsidiary. Such items and the items to which they are made subsidiary are identified in Section G.

G. Items of Work and Construction Details

Bid Item #3 – Seeding

1. This items shall consist of furnishing all the labor, material, equipment, and tools for the preparations of the seedbed, placement of the fertilizer, seeding, and mulch as required in the specified area listed below
2. Seeding areas – including 10ft beyond toe of berm

Seeding Areas			
Description		Quantity	Unit
North Berm and Aux spillway		2.5	acres
South Berm and Aux Spillway		8.5	acres
	Total	11	acres

3. Fertilizer/Mulch

Urea 46-0-0	160 lbs/acre	1,760	lbs
DAP 18-46-0	265 lbs/acre	2,915	lbs
Potash 0-0-60	200 lbs/acre	2,200	lbs
Lime - Ag	2 ton/acre	22 tons	tons
Mulch - straw	none	0	Tons

4. Seed – the seed shall be delivered to the site in tagged and labeled bags showing the percentage purity and germination. The seed shall originate from the United States or Canada, have been tested within 6 months prior to the date of seeding and shall conform to the latest seed and weed laws of Illinois. When the percentage of pure live seed is less than 100%, the rate shall be increased to achieve the same net pure live seed required. Seed shall have a minimum of 80% pure live seed

a. Seeding rate table

Seeding Rate			
Variety	PLS – lbs/acre	Area	Lbs of PLS*
Switchgrass – Cave-in-Rock	10	11	110
Redtop	12	11	132
Oats	20	11	220

**PLS (pure live seed) – Quantities purchased from supplier will be greater than this amount due to % germination and seed purity –*

$$PLS = [(\%germination + \% dormant seed) * \%purity]/100$$

5. The seeding operation shall be performed immediately after notification of contracting officer's technical representative (COTR) has provided inspection of completed earthfill and preparation of the seed bed is complete. Fertilizer, lime, and other soil amendments shall be applied by broadcasting. The fertilizer and soil amendments shall be thoroughly incorporated into the soil immediately following surface application
6. Seed shall be applied uniformly at a depth of ¼ - ½ inch with a drill or cultipacker type seeder. Broadcast methods are acceptable where the seed will be applied uniformly and covered ¼ - ½ inch deep with a cultipacker/cultimulcher, harrow, or similar tool designed to break clods. level, and firm the seedbed.
7. Seedbed shall be inspected by NRCS prior to commencement of seeding. Material certifications shall be provided and reviewed by COTR prior to commencement of seeding. Seeding dates shall be from August 1 to September 10.
8. At any location where the seeding has been displaced by any of the contractor's equipment or personnel, the seeding shall be repaired or replaced immediately at contractor's expense, in a manner acceptable by contracting officer (CO)
9. Compensation shall be according to Method 2 for each type of seeding– under the Measurement and payment section. Payment is by lump sum for seeding.

Construction Specification 21 — Excavation

Title 210 – Engineering

Part 642 – Specifications for Construction Contracts

Subpart B – National Construction Specifications

June 2024

A. Scope

1. The work consists of the excavation required by the drawings and specifications, as well as the disposal of the excavated materials.

B. Classification

1. Excavation is classified as common excavation, rock excavation, or unclassified excavation in accordance with the following definitions.
2. Common excavation is defined as the excavation of all materials that (1) can be excavated, transported, and unloaded using heavy ripping equipment and wheel tractor-scrapers with pusher tractors or (2) can be excavated and dumped into place or loaded onto hauling equipment by excavators having a rated capacity of 1 cubic yard or larger and equipped with attachments (shovel, bucket, backhoe, dragline, or clam shell) appropriate to the material type, character, and nature of the materials.
3. Rock excavation is defined as the excavation of all hard, compacted, or cemented materials that require blasting or the use of ripping and excavating equipment larger than defined for common excavation. The excavation and removal of isolated boulders or rock fragments larger than 1 cubic yard encountered in materials otherwise conforming to the definition of common excavation must be classified as rock excavation. The presence of isolated boulders or rock fragments larger than 1 cubic yard is not in itself sufficient cause to change the classification of the surrounding material.
 - a. For the purpose of these classifications, the following definitions apply:
 - (1) Heavy ripping equipment is a rear-mounted, heavy-duty, single-tooth, ripping attachment mounted on a track-type tractor having a power rating of at least 250-flywheel horsepower, unless otherwise specified in Section J.
 - (2) A wheel tractor-scraper is a self-loading (not elevating) and unloading scraper having a struck-bowl capacity of at least 12 cubic yards.
 - (3) A pusher tractor is a track-type tractor having a power rating of at least 250-flywheel horsepower equipped with appropriate attachments.

4. Unclassified excavation is defined as the excavation of all materials encountered, including rock materials, regardless of their nature or the manner in which they are removed.

C. Blasting

1. The transportation, handling, storage, and use of dynamite and other explosives must be directed and supervised by someone of proven experience and ability who is authorized and qualified to conduct blasting operations.
2. Blasting must be done in a manner that prevents damage to the work or unnecessary fracturing of the underlying rock materials. It must conform to any special requirements in Section J of this specification. When specified in Section J, the contractor must furnish the engineer, in writing, a blasting plan before blasting operations begin.

D. Use of Excavated Material

1. Method 1. To the extent they are needed, all suitable material from the specified excavations must be used in the construction of required permanent earthfill or rockfill. The suitability of material for specific purposes is determined by the engineer. The contractor must not waste or otherwise dispose of suitable excavated material.
2. Method 2. Suitable material from the specified excavations may be used in the construction of required earthfill or rockfill. The suitability of material for specific purposes is determined by the engineer.

E. Disposal of Waste Materials

1. Method 1. All surplus or unsuitable excavated materials are designated as waste and must be disposed of at the locations shown on the drawings.
2. Method 2. All surplus or unsuitable excavated materials are designated as waste and disposed of by the contractor at chosen sites away from the site of the work. The disposal must be in an environmentally acceptable manner that does not violate local rules and regulations.

F. Excavation Limits

3. Excavations must comply with the Occupational Safety and Health Administration Construction Industry Standards Subpart P, Excavations, Trenching, and Shoring (29 CFR pt. 1926). All excavations must be completed and maintained in a safe and stable condition throughout the total construction phase. Structure and trench excavations must be completed to the specified elevations and to the length and width required to safely install, adjust, and remove any forms, bracing, or supports necessary for the installation of the work. Excavations outside the lines and limits shown on the drawings or specified herein are required to meet safety requirements and must be the responsibility of the contractor in constructing and maintaining a safe, stable excavation.

G. Borrow Excavation

1. When the quantities of suitable material obtained from specified excavations are insufficient to construct the specified earthfills and earth backfills, obtain additional material from the designated borrow areas. The extent and depth of borrow pits within the limits of the designated borrow areas are as specified in Section J or as approved by the engineer.
2. Borrow pits must be excavated and finally dressed to blend with the existing topography. They are sloped to prevent ponding and to provide drainage.

H. Over Excavation

1. Correct excavation in rock beyond the specified lines and grades by filling the resulting voids with Portland Cement concrete. Materials and mix proportions must be approved by the engineer. Concrete that will be exposed to the atmosphere when construction is completed must meet the requirements of concrete selected for use under Construction Specification 31, Concrete for Major Structures, or 32, Structure Concrete, as appropriate.
2. Concrete that will be permanently covered must contain not less than five bags of cement per cubic yard. The concrete must be placed and cured as specified by the engineer.
3. Correct excavation in earth beyond the specified lines and grades by filling the resulting voids with approved compacted earthfill. The exception to this is that if the earth will become the subgrade for riprap, rockfill, sand or gravel bedding, or drainfill, the voids may be filled with material conforming to the specifications for the riprap, rockfill, bedding, or drainfill. Before correcting an over excavation condition, the contractor must review the planned corrective action with the engineer and obtain approval of the corrective measures.

I. Measurement and Payment

1. Unit Price Methods. For items of work for which specific unit prices are established in the contract, the volume of each type and class of excavation within the specified pay limits is measured and computed to the nearest cubic yard by the method of average cross-sectional end areas or by methods outlined in Section J of this specification. Regardless of quantities excavated, the measurement for payment is made to the specified pay limits except when excavation outside the specified lines and grades directed by the engineer to remove unsuitable material is included. Excavation required because unsuitable conditions result from the contractor's improper construction operations, as determined by the engineer, is not included for measurement and payment.
2. Method 1. The pay limits are as designated on the drawings.
3. Method 2. The pay limits are defined as follows:
 - a. The upper limit is the original ground surface as it existed before the start of construction operations, unless excavation is performed within areas designated for previous excavation or earthfill; then the upper limit is the modified ground surface resulting from the specified previous excavation or earthfill.

- b. The lower and lateral limits are the neat lines and grades shown on the drawings.
- 4. Method 3. The pay limits are defined as follows:
 - a. The upper limit is the original ground surface as it existed before the start of construction operations, unless excavation is performed within areas designated for previous excavation or earthfill; then the upper limit is the modified ground surface resulting from the specified previous excavation or earthfill.
 - b. The lower and lateral limits is the true surface of the completed excavation as directed by the engineer.
- 5. Method 4. The pay limits are defined as follows:
 - a. The upper limit is the original ground surface as it existed before the start of construction operations, unless excavation is performed within areas designated for previous excavation or earthfill; then the upper limit is the modified ground surface resulting from the specified previous excavation or earthfill.
 - b. The lower limit is at the bottom surface of the proposed structure.
 - c. The lateral limits are 18 inches outside of the outside surface of the proposed structure or are the vertical planes 18 inches outside of and parallel to the footings, whichever gives the larger pay quantity, except as provided in paragraph d below.
 - d. For trapezoidal channel linings or similar structures that are to be supported by the sides of the excavation without intervening forms, the lateral limits are at the underside of the proposed lining or structure.
 - e. For the purposes of the definitions in paragraphs b, c, and d above, any specified bedding or drainfill directly beneath or beside the structure will be considered to be a part of the structure.
- 6. All Methods. The following provisions apply to all methods of measurement and payment.
 - a. Payment for each type and class of excavation is made at the contract unit price for that type and class of excavation. Such payment will constitute full compensation for all labor, materials, equipment, and other items necessary and incidental to the performance of the work with the exception that extra payment for backfilling over excavation will be made in accordance with the following provisions.
 - b. Payment for backfilling over excavation, as specified in Section H of this specification, is made only if the excavation outside specified lines and grades is directed by the engineer to remove unsuitable material and the unsuitable condition is not a result of the contractor's improper construction operations as determined by the engineer.
 - c. Compensation for any item of work described in the contract but not listed in the bid schedule is included in the payment for the item of work to which it is made subsidiary. Such items and the items to which they are made subsidiary are identified in Section J of this specification.

J. Items of Work and Construction Details

Bid Item #3 – Excavation – 12,650 cubic yards

North Berm Stripping	1,584 cy	Depth of 0.5ft
North Pool Stripping	1,375 cy	Depth of 0.5ft
North Aux Spillway	205 cy	As shown in plans
South Berm Stripping	4,830 cy	Depth of 0.5ft
South Pool Stripping	3,810 cy	Depth of 0.5ft
South Aux Spillway	846 cy	As shown in plans

Item of work to be performed in conformance with this specification and construction details are

7. This item shall consist of common excavation required to strip the north and south berm footprints and north and south wetland pool area of the top 6” of vegetative material as shown in drawing and specifications.
8. In section E – Disposal of material – use section 1, with undesirable material with vegetation or rooting material shall be placed as follows:
 - a. The stripped material from under the footprint of the berm can be used as the top 6” of berm fill consistent with topsoiling. Material can temporarily be stockpiled for placement on berm
 - b. The striped material from the surface of the wetland pools shall be placed back in the wetland pool after material is removed for earthfill. The stripped material shall be placed back in the wetland pool area with approximately 6” depth. Irregular surface is adequate with tolerance of +/- 0.5ft
 - c. See earthfill specification for allowable material
9. Under Section I – Measure and Payment
 - a. No payment method shall be used. Payment for each type of excavation is made at the contract unit price for that type of excavation. Such payment will constitute full compensation for all labor, material, equipment, and all other items necessary and incidental to the performance of the work, with the exception that the final quantities shall not be measured unless there is a significant change to the plans. It shall be the contractor's responsibility to prove this quantity difference to the COTR. If there are no changes to the plans, then the estimated volumes shall be used

Construction Specification 45 — Plastic Pipe

Title 210 – Engineering

Part 642 – Specifications for Construction Contracts

Subpart B – National Construction Specifications

June 2024

A. Scope

1. The work consists of furnishing and installing plastic pipe (except corrugated polyethylene pipe) and the necessary fittings and appurtenances as shown on the drawings or as specified herein.

B. Material

1. Pipe, fittings, and gaskets must conform to the requirements of Material Specification 547, Plastic Pipe, and as specified in Section N of this specification or as shown on the drawings.
2. Perforated pipe must conform to the requirements of Material Specification 547, Plastic Pipe, and as specified in Section N of this specification or as shown on the drawings.
3. Unless otherwise specified, concrete must conform to the requirements of Construction Specification 32, Structure Concrete, and Section H of this specification.
4. Unless otherwise specified, earth backfill must conform to the requirements of Construction Specification 23, Earthfill.
5. Unless otherwise specified, drainfill must conform to the requirements of Construction Specification 24, Drainfill.

C. Handling and Storage

1. Deliver pipe to the job site and handled by means that provide adequate support to the pipe and do not subject it to undue stresses or damage. When handling and placing plastic pipe, take care to prevent impact blows, abrasion damage, and gouging or cutting by metal edges, surfaces or rocks. The manufacturer's special handling requirements must be strictly observed. Special care must be taken to avoid impact when the pipe must be handled at a temperature of 40 degrees Fahrenheit or less.
2. Store pipe on a relatively flat surface so that the barrels are evenly supported. Unless the pipe is specifically manufactured to withstand exposure to ultraviolet radiation, cover it with an opaque material when stored outdoors for 15 days or longer.

D. Excavation

1. Excavate in accordance with Construction Specification 21, Excavation, and Section N of this specification or as shown on the drawings.
2. Excavate the pipe foundation a minimum of 4 inches lower than the pipe grade shown on the drawings or staked in the field whenever bedrock, boulders, cobbles, or other material that may cause pipe damage is encountered at planned pipe grade.

E. Laying the Pipe

1. Install plastic pipe conduits complete with fittings and other related appurtenances to the lines and grades shown on the drawings or specified in Section N of this specification. Install the pipe so that there is no reversal of grade between joints unless otherwise shown on the drawings. The pipe must not be dropped or dumped on the bedding or into the pipe trench. The ground surface near the pipe trench must be free of loose rocks and stones greater than 1 inch in diameter. This ensures that rock will not be displaced and impact the pipe.
2. Just before placement, inspect each pipe section to ensure that all foreign material is removed from inside the pipe. The pipe ends and the couplings must be free of foreign material when assembled. At the completion of a work shift, all open ends of the pipeline must be temporarily closed off using a suitable cover or plug.
3. Take care to prevent distortion and damage during hot or cold weather. During unusually hot weather (daytime high temperature of more than 90 °F), lightly backfill the pipe assembled in the trench or shade to keep it as near to ground temperature as possible until final backfill is placed. Backfill operations must be performed during daily construction periods when the ground temperature and the temperature of the pipe do not vary more than 40 degrees Fahrenheit.
4. Install perforated pipe with the perforations down and oriented symmetrically about the vertical centerline. Perforations must be clear of any obstructions on the inside and outside of the pipe when the pipe is approved by the engineer for backfill.
5. During installation, firmly and uniformly bed the pipe throughout its entire length to the depth and in the manner specified in Section N of this specification or as shown on the drawings. Place bell-holes in bedding material under bells, couplings, and other fittings to ensure the pipe is uniformly supported throughout its entire length. Blocking or mounding beneath the pipe to bring the pipe to final grade is not permitted.

F. Pipe Embedment

1. Earth bedding.
 - a. Firmly and uniformly place the pipe on compacted earthfill bedding or an in-place earth material bedding of ample bearing strength to support the pipe without noticeable settlement. The earth material on which the pipe is placed must have uniform density to prevent differential settlement.

- b. Unless otherwise specified, form a groove that closely conforms to the outside surface of the pipe in the bedding. The depth of the groove must be equal to or greater than 0.3 of the pipe diameter.
 - c. Compact earth bedding to a density not less than adjacent, undisturbed, in-place earth material or be compacted earth backfill. Earthfill material used for compacted earth bedding must be free of rocks or stones greater than 1 inch in diameter and earth clods greater than 2 inches in diameter. The pipe must be loaded sufficiently during the compaction of bedding under the haunches and around the sides of the pipe to prevent displacement from its final approved placement.
- 2. Sand, gravel, or crushed rock bedding. When sand, gravel, or crushed rock bedding is specified, firmly and uniformly place the pipe on the bedding material. Material for bedding must not exceed 1 inch in diameter. Unless otherwise specified in Section N of this specification or shown on the drawings, the coarse-grained bedding material must be carefully placed and compacted to a depth equal to or greater than 0.3 of the diameter of the pipe above the bottom of the pipe. The pipe must be loaded sufficiently during backfilling and compaction around the sides to prevent displacement of the pipe from its final approved placement.
 - 3. Pipe encased in drainfill. Firmly and uniformly place the pipe on bedding of specified drainfill, which must be placed and compacted as specified in Section N of this specification or as shown on the drawings to form a continuous, uniform support around the entire circumference of the pipe. The pipe must be loaded sufficiently during backfilling around the sides and during compaction to prevent displacement of the pipe.

G. Backfill

- 1. Initial backfill.
 - a. Initial backfill must be placed in two stages. In the first stage (haunching), backfill is placed to the pipe spring line (center of pipe). In the second stage, it is placed to 6 inches above the top of the pipe.
 - b. The first stage material must be worked carefully under the haunches of the pipe to provide continuous support throughout the entire pipe length. The haunching backfill material must be placed in layers that have a maximum thickness of about 6 inches and are compacted as specified in Section N of this specification or as shown on the drawings. During compaction operations, care must be taken to ensure that the tamping or vibratory equipment does not come in contact with the pipe, and the pipe is not deformed or displaced.
 - c. When pressure testing is not specified, cover the pipe with a minimum of 6 inches of backfill material as soon as possible following assembly of the pipe in the trench but not later than within the same day that placement has occurred. When pressure testing is specified, place sufficient backfill material over the pipe to anchor the conduit against movement during pressure testing activities.
- 2. Final backfill.

- a. Final backfill consists of placing the remaining material required to complete the backfill from the top of the initial backfill to the ground surface, including mounding at the top of the trench. Final backfill material within 2 feet of the top of the pipe must be free of debris or rocks larger than 3 inches nominal diameter. Coarse backfill material must be the specified sand, gravel, crushed rock, or drainfill. Final backfill must be placed in approximately uniform, compacted layers. Final backfill compaction requirements must be as specified in Section N of this specification or as shown on the drawings.
- b. Vehicles or construction equipment must not be allowed to cross the pipe until the minimum earth cover and required density as specified in Section 14 of this specification have been obtained.

H. Joints

1. Unless otherwise specified in Section N of this specification or shown on the drawings, joints must be either bell-and-spigot types with elastomeric gaskets, coupling types, solvent cement bell and spigot, or jointed by butt heat fusion. When a lubricant is required to facilitate joint assembly, it must be a type having no deleterious effect on the gasket or pipe material.
2. Pipe joints must be watertight at the pressures specified except where unsealed joints are indicated.
3. Install pipe and join in accordance with the manufacturer's recommendations. Laying deflections and joint fitting or stab depths must be within the manufacturer's recommended tolerances.
4. When solvent cement joints are specified for PVC or ABS pipe and fittings, they must be made in accordance with the following ASTMs and the related appendix of each ASTM: D2855 for PVC pipe and fittings and D2235 for ABS pipe and fittings.
5. Flanged, banded, heat-fusion, or elastomeric-sealed mechanical joints must be used when joining polyethylene (PE) and high-density polyethylene (HDPE) pipe and fittings unless otherwise specified in Section N of this specification or as shown on the drawings.
6. Cut pipe ends square and debur to provide a uniform, smooth surface for the jointing process. Place reference marks on the spigot ends to assist in determining when proper seating depth has been achieved within the joint.

I. Fittings

1. Unless otherwise specified, steel fittings, valves, and bolted connections must be painted or coated as recommended by the manufacturer.
2. Fittings for non-pressure pipe must be of the same or similar material as the pipe and must provide the same durability, watertightness, and strength as the pipe unless otherwise specified.

J. Pressure Testing

1. Method 1. Pressure testing of the completed conduit is not required.

K. Measurement and Payment

1. Method 1. For items of work for which specific unit prices are established in the contract, the quantity of each kind, size, and class of pipe is determined to the nearest foot by measurement of the laid length along the crown centerline of the conduit. Payment for each kind, size, and class of pipe is made at the contract unit price for that kind, size, and class. Such payment constitutes full compensation for furnishing, transporting, and installing the pipe including excavation, shoring, backfill, bedding, thrust blocks, and all fittings, appurtenances, and other items necessary and incidental to the completion of the work. Payment for appurtenances listed separately in the bid schedule is made at the contract prices for those items.
2. Method 2. For items of work for which specific unit prices are established in the contract, the quantity of each kind, size, and class of pipe is determined as the sum of the nominal laying lengths of the sections used. Payment for each kind, size, and class of pipe is made at the contract unit price for the kind, size, and class. Such payment constitutes full compensation for furnishing, transporting, and installing the pipe including excavation, shoring, backfill, bedding, thrust blocks, and all fittings, appurtenances, and other items necessary and incidental to the completion of the work. Payment for appurtenances listed separately in the bid schedule is made at the contract prices for those items.
3. Method 3. For items of work for which specific lump sum price are established in the contract, the quantity of each kind, size, and class of pipe is determined to the nearest foot by measurement of the laid length along the crown centerline of the conduit. Payment for each kind, size, and class of pipe is made at the contract lump sum price. Such payment constitutes full compensation for furnishing, transporting, and installing the pipe including shoring, all fittings, thrust blocks, appurtenances, and other items necessary and incidental to the completion of the work. Payment for appurtenances listed separately in the bid schedule is made at the contract prices for those items.
4. Method 4. For items of work for which specific unit prices are established in the contract, the quantity of each kind, size, and class of pipe is determined as the sum of the nominal laying lengths of the pipe sections used. Payment for each kind, size, and class of pipe is made at the contract unit price for that kind, size, and class. Such payment constitutes full compensation for furnishing, transporting, and installing the pipe including shoring, all fittings, thrust blocks, appurtenances, and other items necessary and incidental to the completion of the work. Payment for appurtenances listed separately in the bid schedule is made at the contract prices for those items.
5. Methods 3 and 4. Excavation, backfill, and bedding are paid separately under their respective bid items.
6. All Measurement and Payment Methods. Compensation for any items of work described in the contract but not listed in the bid schedule is included in the payment for the item of work to which it is made subsidiary. Such items and items to which they are made subsidiary are identified in Section N of this specification.

L. Items of Work and Construction Details

Bid Item #4 – Plastic pipe and Water Control Structures

1. Material as shown on construction plans

<u>Location</u>	<u>Length</u>	<u>Size/Specification</u>
North Pipe	100ft with anti-seep collar	As shown on plans
North WCS	1 inline water control str	As shown on plans
South Pipe	100ft with anti-seep collar	As shown on plans
South WCS	1 inline water control str	As shown on plans

2. The materials and manufacture of water control structures, pipe, coupling bands, coatings, and other appurtenances must be as shown on the construction plans and conform to the NRCS material specifications. Place water control structure and pipe couplers on a stable base. The base may be compacted earth or concrete pad. Extend the stable base no less than 1 foot around structure. Install structure with all stop boards in tracks. Place impervious backfill material around the structure and other appurtenances by hand and in layers in accordance with earthfill specification. Achieve proper compaction around structure without the use of mechanical compaction. Increase height of fill at approximately the same rate on all sides of the structure. Heavy equipment must not be operated within 2 feet of any structure. Lay the pipe to the lines, grades, and elevations shown on the drawings and bed the pipes firmly and uniformly throughout their entire length. Install the pipe according to the manufacturer's instructions. Select anti-seep collars composed of materials compatible with that of the pipe and install anti-seep collars so that they are watertight. Install pipe with the outside laps of circumferential joints pointing upstream and with longitudinal laps on the sides at approximately the vertical mid-height of the pipe. Field welding of corrugated galvanized steel pipe will not be permitted. Join pipe sections with coupling bands or other approved methods, as shown on the construction drawings. All pipe connections must be watertight.
3. The following reference specifications pertain to conduit materials currently acceptable for use.

REFERENCE SPECIFICATIONS FOR CONDUIT MATERIALS	
Description	ASTM
Plastic	
Corrugated Polyethylene (PE) Pipe and Fittings	F405 F667
Poly Vinyl Chloride (PVC) Pipe and Fittings	F949 D1785 D2241
Styrene-Rubber (SR) Plastic Drain Pipe and Fittings	D2852
Dual Wall Polyethylene Pipe	
Corrugated Polyethylene (PE) Pipe and Fittings	F2306 F2648 F405 F667
Elastomeric Seals and Joints (Gaskets)	F477 D3212
Clay	
Clay Drain Tile and Pipe	C4 C700 C301
Concrete	
Concrete Drain Tile and Pipe	C412 C118 C14 C76 C444
Test Methods for Concrete Pipe	C497
Portland Cement	C150
Metal	
Corrugated Aluminum Pipe	B745
Corrugated Steel Pipe	A760

4. Take special care during backfill operations not to disturb the grade and alignment. Tie down or load pipe sufficiently during backfilling around the

sides to prevent its being lifted from the bedding. Backfill material will have sufficient moisture so that optimum compaction can be obtained. Add water to backfill as necessary to achieve desired moisture content. Place backfill around the pipe and anti-seep collars in layers not more than 4 inches thick before compaction. Compact each layer of backfill with power tampers, hand tampers, or plate vibrators to the same density requirements as specified for the adjacent embankment. Select compaction method based on soil type. Bring up backfill over and around the pipe and anti-seep collar uniformly on all sides. The passage of earth moving equipment will not be allowed over the pipe or anti-seep collar until backfill has been placed above the top of the pipe or collar to a depth of two (2) feet. Hand tamping, only, should be used to compact the fill under the bottom half of circular pipes. Do not operate equipment weighing 400 pounds or more per foot of width within 2 feet of any structure or pipe.

5. Animal Guards

- a. All inlet and outlet pipes must have trash rack and animal guards, as shown on the construction drawings. Guards on outlet pipe must allow passage of debris. Install the trash rack as shown in plans. Install animal guards to manufacturer specifications

6. Measurement and Payment

- a. Shall be Method, expect will be payment by lump sum instead of per unit. Installed material will require a material specification certification and quantity documentation of what was installed. All material certificates and quantity documentation will need to be provided to COR.

Construction Specification 23 — Earthfill

Title 210 – Engineering

Part 642 – Specifications for Construction Contracts

Subpart B – National Construction Specifications

June 2024

A. Scope

1. The work consists of the construction of earth embankments, other earthfills, and earth backfills required by the drawings and specifications.
2. Earthfill is composed of natural earth materials that can be placed and compacted by construction equipment operated conventionally.
3. Earth backfill is composed of natural earth material placed and compacted in confined spaces or adjacent to structures (including pipes) by hand tamping, manually directed power tampers or vibrating plates, or their equivalent.

B. Material

1. All fill material must be obtained from required excavations and designated borrow areas. The selection, blending, routing, and disposition of material in the various fills must be subject to approval by the engineer.
2. Fill materials must contain no frozen soil, sod, brush, roots, or other perishable material. Rock particles larger than the maximum size specified for each type of fill must be removed prior to compaction of the fill.
3. The types of material used in the various fills must be as listed and described in the specifications and drawings.

C. Foundation preparation

1. Foundations for earthfill must be stripped to remove vegetation and other unsuitable material or must be excavated as specified.
2. Except as otherwise specified, earth foundation surfaces must be graded to remove surface irregularities and must be scarified parallel to the axis of the fill or otherwise acceptably scored and loosened to a minimum depth of 2 inches. The moisture content of the loosened material must be controlled as specified for the earthfill and the surface material of the foundation must be compacted and bonded with the first layer of earthfill as specified for subsequent layers of earthfill.
3. Earth abutment surfaces must be free of loose, uncompacted earth in excess of 2 inches in depth normal to the slope and must be at such a moisture content that the earthfill can be compacted against them to produce a good bond between the fill and the abutments.
4. Rock foundation and abutment surfaces must be cleared of all loose material by hand or other effective means and must be free of standing water when fill is

placed upon them. Occasional rock outcrops in earth foundations for earthfill, except in dams and other structures designed to restrain the movement of water, must not require special treatment if they do not interfere with compaction of the foundation and initial layers of the fill or the bond between the foundation and the fill.

5. Foundation and abutment surfaces must be no steeper than one horizontal to one vertical unless otherwise specified. Test pits or other cavities must be filled with compacted earthfill conforming to the specifications for the earthfill to be placed upon the foundation.

D. Placement

1. Earthfill must not be placed until the required excavation and foundation preparation have been completed and the foundation has been inspected and approved by the engineer. Earthfill must not be placed upon a frozen surface nor have snow, ice, or frozen material incorporated in the earthfill matrix.
2. Earthfill must be placed in approximately horizontal layers. The thickness of each layer before compaction must not exceed the maximum thickness specified in section J or shown on the drawings. Materials placed by dumping in piles or windrows must be spread uniformly without exceeding the specified thickness before being compacted.
3. Hand-compacted earth backfill must be placed in layers whose thickness before compaction does not exceed the maximum thickness specified for layers of earth backfill compacted by manually directed power tampers.
4. Place earth backfill in a manner that prevents damage to the structures and allows the structures to assume the loads from the earth backfill gradually and uniformly. The height of the earth backfill adjacent to a structure must be increased at approximately the same rate on all sides of the structure.
5. Meet the following additional requirements for earthfill and earth backfill in dams, levees, and other structures designed to restrain the movement of water.
 - a. The distribution of materials throughout each zone must be essentially uniform and the earthfill free from lenses, pockets, streaks, or layers of material differing substantially in texture, moisture content, or gradation from the surrounding material. Zone earthfills must be constructed concurrently unless otherwise specified.
 - b. Scarified the surface of each layer parallel to the axis of the fill to at minimum a depth of 2 inches before the next layer is placed.
 - c. The top surface of embankments must be maintained approximately level during construction with two exceptions: a crown or cross-slope of about 2 percent must be maintained to ensure effective drainage, or as otherwise specified for drainfill or sectional zones.
 - d. Construct dam embankments in continuous layers from abutment to abutment except where openings to facilitate construction or to allow the passage of streamflow during construction are specifically authorized in the contract.

- e. Embankments built at different levels as described under paragraphs c or d above must be constructed so that the slope of the bonding surfaces between embankments in place and embankments to be placed is not steeper than 3 feet horizontal to 1 foot vertical. The bonding surface of the embankment in place must be stripped of all material not meeting the requirements of this specification and must be scarified, moistened, and recompactd when the new earthfill is placed against it. This ensures a good bond with the new earthfill and obtains the specified moisture content and density at the contact of the in-place and new earthfills.

E. Control of moisture content

1. During placement and compaction of earthfill and earth backfill, the moisture content of the material being placed must be maintained within the specified range.
2. The application of water to the earthfill material must be accomplished at the borrow areas insofar as practicable. Water may be applied by sprinkling the material after placement on the earthfill, if necessary. Obtain a uniform moisture distribution by disking.
3. Material that is too wet when deposited on the earthfill must either be removed or dried to the specified moisture content prior to compaction.
4. If the top surface of the preceding layer of compacted earthfill or a foundation or abutment surface in the contact zone with the earthfill becomes too dry to permit suitable bond, it must either be removed or scarified and moistened by sprinkling to an acceptable moisture content before placement of the next layer of earthfill.
5. The moisture content of the earthfill shall be such that a ball will not readily separate when made in the hand or will easily ribbon out between the thumb and finger. Moisture content will be considered too high for earthfill operations when conditions will not allow for efficient equipment and compaction operations. Conditions should not allow the pumping or bulging of soil under weight of compaction equipment. Contractor is responsible for controlling moisture during fill operations at optimum conditions as explained above

F. Compaction

1. Earthfill. Earthfill must be compacted according to the following requirements for the class of compaction specified:
 - a. Class C compaction. Compact each layer of earthfill by the specified number of passes of the type and weight of roller or other equipment specified or by an equivalent approved method. Each pass must consist of at least one passage of the roller wheel or drum over the entire surface of the layer.
2. Earth backfill. Compact earth backfill adjacent to structures to a density equivalent to that of the surrounding in-place earth material or adjacent required earthfill or earth backfill. Compaction must be accomplished by hand tamping or manually directed power tampers, plate vibrators, walk-behind, miniature, or self-

propelled rollers. Unless otherwise specified, heavy equipment, including backhoe-mounted power tampers or vibrating compactors, and manually directed vibrating rollers must not be operated within 3 feet of any structure. Towed or self-propelled vibrating rollers must not be operated within 5 feet of any structure. Compaction by means of drop weights operating from a crane or hoist is not permitted.

3. Heavy Equipment. The passage of heavy equipment will not be allowed:
 - a. Over cast-in-place conduits within 14 days after placement of the concrete
 - b. Over cradled or bedded precast conduits within 7 days after placement of the concrete cradle or bedding
 - c. Over any type of conduit until the backfill has been placed above the top surface of the structure to a height equal to one-half the clear span width of the structure or pipe or 3 feet, whichever is greater, except as may be specified in Section J.
4. Compacting adjacent to structures. Compacting of earth backfill adjacent to structures must not be started until the str is proper set. Minimum of 2ft clearance from structure or top of pipe require hand compaction.

G. Reworking or removal and replacement of defective earthfill

1. Earthfill placed at densities lower than the specified minimum density or at moisture contents outside the specified acceptable range of moisture content, or otherwise not conforming to the requirements of the specifications must be reworked to meet the requirements or removed and replaced by acceptable earthfill. The replacement earthfill and the foundation, abutment, and earthfill surfaces upon which it is placed must conform to all requirements of this specification for foundation preparation, approval, placement, moisture control, and compaction.

H. Testing

1. During the course of the work, the contractor must perform quality control tests, as applicable, to identify earthfill and earth backfill materials; determine the reference maximum density and optimum moisture content; and document that the moisture content of material at the time of compaction and the density of earthfill and earth backfill in place conform to the requirements of this specification.
2. Correction for Oversize Particles. If the materials to be used for earthfill or earth backfill contain more than 5 percent by dry weight of oversize rock particles (particles larger than those allowed in the specified compaction test and method), corrections for oversize particles must be made using the appropriate procedures explained in ASTM D4718.

I. Measurement and payment

1. Unit Price Methods. For items of work for which specific unit prices are established in the contract, the volume of each type and compaction class of earthfill and earth backfill within the specified zone boundaries and pay limits is

measured and computed to the nearest cubic yard by the method of average cross-sectional end areas. Unless otherwise specified in Section J, no deduction in volume is made for embedded items such as, but not limited to, conduits, inlet structures, outlet structures, embankment drains, sand diaphragm and outlet, and their appurtenances.

2. Pay Limits. The pay limits must be as defined below, with the further provision that earthfill required to fill voids resulting from over excavation of the foundation, outside the specified lines and grades, will be included in the measurement for payment only under the following conditions:
 - a. Where such over excavation to remove unsuitable material is directed by the engineer
 - b. Where the unsuitable condition is not a result of the contractor's improper construction operations as determined by the engineer.
3. Beyond Specified Lines and Grades. Earthfill beyond the specified lines and grades to backfill excavation required for compliance with OSHA requirements will be considered subsidiary to the earthfill bid items.
4. Method 1. The pay limits must be as designated on the drawings.
5. Method 2. The pay limits must be the measured surface of the foundation when approved for placement of the earthfill and the specified neat lines of the earthfill surface.
6. Method 3. The pay limits must be the measured surface of the foundation when approved for placement of the earthfill and the measured surface of the completed earthfill.
7. Method 4. The pay limits must be the specified pay limits for excavation and the specified neat lines of the earthfill surface.
8. Method 5. The pay limits must be the specified pay limits for excavation and the measured surface of the completed earthfill.
9. Method 6. Payment for each type and compaction class of earthfill and earth backfill is made at the contract unit price for that type and compaction class of earthfill. Such payment will constitute full compensation for all labor, material, equipment, and other items necessary and incidental to the performance of the work.
10. Method 7. Payment for each type and compaction class of earthfill and earth backfill is made at the contract unit price for that type and compaction class of earthfill. Such payment will constitute full compensation for all labor, material, equipment, and other items necessary and incidental to the performance of the work except furnishing, transporting, and applying water to the foundation and earthfill material. Water applied to the foundation and earthfill material is measured and payment made as specified in Construction Specification 10.
11. All methods. The following provisions apply to all methods of measurement and payment. Compensation for any item of work described in the contract but not listed in the bid schedule is included in the payment for the item of work to which

it is made subsidiary. Such items and the items to which they are made subsidiary are identified in Section J of this specification.

J. Items of work and construction details

Bid Item #5 – Earthfill – 21,149 cubic yards

12. This item shall consist of placement of earthfill for the designated berm area.

North Berm	4,310cy + 1,584cy topsoil = 5,894cy
South Berm	10,425cy + 4,830 topsoil = 15,255cy

13. All borrow material shall come from wetland pool locations after vegetative material is stripped in accordance with Excavation Specification 21

14. Topsoiling, construction specification 3, is subsidiary to this specification and is included in bid of earthfill. The stripped excavated material from the footprint of each berm can be used on the top 6” of the berm as topsoil

15. Construction tolerance for the top of the berm is -0.0 and +0.2. Top width and side slopes with 1.0ft of shown on construction plans

16. Compaction requirements

- a. Max lift thickness is 9” for machine equipment. A minimum of 2 complete full width passed required with sheepfoot style compaction equipment with minimum of 200PSI rating or till feet walk out of soil by 50% of foot height.
- b. Max lift thickness for hand compaction is 4” thickness. The location of hand compaction is around pipe and structure. Compaction equipment shall be jumping jack style compaction equipment. NRCS inspection is required with placing earthfill under pipe haunches and around pipe and structure.

17. Measurement and Payment – shall be by Method 6 with the exception that the final quantities shall not be measured unless there is a significant change to the plans. It shall be the contractor's responsibility to prove this quantity difference to the COTR. If there are no changes to the plans, then the estimated volumes shall be used.

Construction Specification 3 — Structure Removal

Title 210 – Engineering

Part 642 – Specifications for Construction Contracts

Subpart B – National Construction Specifications

June 2024

A. Scope

1. The work consists of the removal, salvage, and disposal of structures (including subsurface drains from the designated areas.

B. Marking

1. Method 1. Each structure or structure part to be removed will be marked with stakes, flags, paint, or other suitable methods.
2. Method 2. The area boundaries from which structures must be removed will be marked using stakes, flags, paint, or other suitable methods. Structures to remain undisturbed or to be salvaged will be designated by special markings.

C. Removal

1. Method 1. Remove all structures designated for removal in the contract to the specified extent and depth.
2. Method 2. Within the areas so marked, remove all visible and buried structures identified to the specified extent and depth.

D. Salvage

1. Structures or structure parts that are designated to be salvaged must be carefully removed and neatly placed in the specified or approved storage location. Salvaged structures that are capable of being disassembled must be dismantled into individual members or sections. Such structures must be neatly and systematically matchmarked with paint before disassembly. Mark all connectors and other parts to indicate their proper location within the structure and fasten to the appropriate structural member or packed in suitable containers.
2. Material from fences designated to be salvaged must be placed outside the work area on the property that the fence was originally located. Fence wire must be rolled into uniform rolls of suitable size and neatly piled with other salvaged materials. Neatly stack posts and rails.

E. Disposal of Refuse Materials

1. Dispose of refuse materials resulting from structure removal in a manner and at locations specified in Section G of this specification or in an acceptable manner and at locations approved by the contracting officer. Disposal by burning must be in accordance with local rules and regulations.

F. Measurement and Payment

1. Method 1.
 - a. For items of work for which specific unit prices are established by the contract, payment for the removal of each structure unit, except fences, is made at the contract unit price. Fences removed or removed and salvaged are measured to the nearest linear foot. Payment for fence removal or removal and salvage is made at the contract unit prices for each type and size of fence.
 - b. Such payment will constitute full compensation for all labor, equipment, tools, applicable permits and associated fees for burning and disposal of refuse, and other items necessary and incidental to the completion of the work.
2. Method 2.
 - a. For items of work for which specific lump-sum prices are established by the contract, payment for structure removal is made at the contract lump-sum price.
 - b. Such payment will constitute full compensation for all labor, equipment, tools, applicable permits and associated fees for burning and disposal of refuse, and other items necessary and incidental to the completion of the work.
3. All Methods. The following provisions apply to all methods of measurement and payment. Compensation for any item of work described in the contract but not listed as a contract line-item number in the bid schedule is included in the payment for the item of work to which it is made subsidiary. Such items and items to which they are made subsidiary are identified in Section G of this specification.

G. Items of Work and Construction Details

Bid Item #6 - Tile Decommissioning – 750ft

1. This item shall consist of structure removal and disposal, an trench is required to expose existing tile as shown in location(s) on the plan. The contractor will be responsible for exploration and finding of drains in addition to the dimensions shown in the plans
2. The exploration trench shall be inspected by NRCS prior to backfill. Contractor is to coordinate timing to ensure NRCS inspector is present during excavation if contractor wants to backfill immediately after being dug
3. Exploration trench requirements
 - a. Excavated to a min depth of 6'
 - b. Trench shall be immediately backfilled in all locations where no drains are found after NRCS inspection or shall have side slopes of trench sloped back or benched to meet OSHA requirements.
 - c. Backfill compaction only required under footprint of berm. Shall meet the requirements of 6" lifts and tamped to firmness of surrounding

natural ground. Backfill area outside berm footprint does not require compaction but needs mounded for settlement.

- d. In event of located drain
 - i. Area shall not be backfilled until inspected by NRCS. OSHA trench requirements will need to be comply
 - ii. NRCS shall be notified of all location's drains are found
 - iii. Contractor shall provide NRCS with the following information
 - 1. Location and/or station of drain, Size, material and flow direction of drain.
 - iv. If determined to drain adjacent property, NRCS will provide additional decommissioning instruction in accordance with IL drainage law.
 - v. If tile is found to not drain adjacent land, then contractor is to excavate drain 100' downstream and 200' upstream or as shown dimensions on plans
- 4. Disposal of refuse material
 - a. Contractor is to dispose of refuse material offsite or by approved NRCS methods.
- 5. Method of Payment
 - a. Method 2 – but the additional exploration trenching to find the drain are included lump sum payment.

Construction Specification 61 — Rock Riprap

Title 210 – Engineering

Part 642 – Specifications for Construction Contracts

Subpart B – National Construction Specifications

June 2024

A. Scope

1. The work consists of the construction of rock riprap revetments and blankets, including filter or bedding where specified.

B. Material

1. Rock riprap must conform to the requirements of Material Specification 523, Rock for Riprap, or, if so specified, must be obtained from designated sources. It must be free from dirt, clay, sand, rock fines, and other material not meeting the required gradation limits.
2. At least 10 days before rock is delivered from other than designated sources, the contractor must designate in writing the source from which rock material will be obtained and provide information satisfactory to the contracting officer that the material meets contract requirements. The contractor must provide the responsible engineer free access to the source for the purpose of obtaining samples for testing. The size and grading of the rock must be as specified in section H.
3. Rock from approved sources must be excavated, selected, and processed to meet the specified quality and grading requirements at the time the rock is installed.
4. Filter or bedding aggregates, when required, must conform to Material Specification 521, Aggregates for Drainfill and Filters, unless otherwise specified. Geotextiles must conform to Material Specification 592, Geotextile.

C. Subgrade preparation

1. The subgrade surface on which the rock riprap, filter, bedding, or geotextile is to be placed must be cut or filled and graded to the lines and grades shown on the drawings. When fill to subgrade lines is required, it must consist of approved material and conform to the requirements of the specified class of earthfill.
2. Rock riprap, filter, bedding, or geotextile must not be placed until the foundation preparation is completed and the subgrade surface has been inspected and approved.

D. Equipment-placed rock riprap

1. The rock riprap must be placed by equipment on the surface and to the depth specified. It must be installed to the full course thickness in one operation and in such a manner as to avoid serious displacement of the underlying material. The

rock for riprap must be delivered and placed in a manner that ensures the riprap in place is reasonably homogeneous with the larger rocks uniformly distributed and firmly in contact one to another with the smaller rocks and spalls filling the voids between the larger rocks. Some hand placing may be required to provide a neat and uniform surface.

2. Rock riprap must be placed in a manner to prevent damage to structures. Hand placing is required as necessary to prevent damage to any new and existing structures.

E. Hand-placed rock riprap

1. The rock riprap must be placed by hand on the surface and to the depth specified. It must be securely bedded with the larger rocks firmly in contact one to another without bridging. Spaces between the larger rocks must be filled with smaller rocks and spalls. Smaller rocks must not be grouped as a substitute for larger rock. Flat slab rock must be laid on its vertical edge except where it is laid like paving stone and the thickness of the rock equals the specified depth of the riprap course.

F. Filter or bedding

1. When the contract specifies filter, bedding, or geotextile beneath the rock riprap, the designated material must be placed on the prepared subgrade surface as specified. Compaction of filter or bedding aggregate is not required, but the surface of such material must be finished reasonably smooth and free of mounds, dips, or windrows.

G. Measurement and payment

1. Method 1.
 - a. For items of work for which specific unit prices are established in the contract, the quantity of each type of rock riprap placed within the specified limits is computed to the nearest ton by actual weight. The volume of each type of filter or bedding aggregate is measured within the specified limits and computed to the nearest cubic yard by the method of average cross-sectional end areas. For each load of rock riprap placed as specified, the contractor must furnish to the responsible engineer a statement-of-delivery ticket showing the weight to the nearest 0.1 ton.
 - b. Payment is made at the contract unit price for each type of rock riprap, filter, or bedding. Such payment is considered full compensation for completion of the work.
2. Method 2.
 - a. For items of work for which specific unit prices are established in the contract, the quantity of each type of rock riprap placed within the specified limits is computed to the nearest 0.1 ton by actual weight. The quantity of each type of filter or bedding aggregate delivered and placed within the specified limits is computed to the nearest 0.1 ton. For each load of rock riprap placed as specified, the contractor must furnish to the engineer a statement-of-delivery ticket showing the weight to the nearest 0.1 ton. For each load of filter or

bedding aggregate, the contractor must furnish to the responsible engineer a statement-of-delivery ticket showing the weight to the nearest 0.1 ton.

- b. Payment is made at the contract unit price for each type of rock riprap, filter, or bedding. Such payment is considered full compensation for completion of the work.

3. Method 3.

- a. For items of work for which specific unit prices are established by the contract, the volume of each type of rock riprap and filter or bedding aggregate is measured within the specified limits and computed to the nearest cubic yard by the method of average cross-sectional end areas.
- b. Payment is made at the contract unit price for each type of rock riprap, filter, or bedding. Such payment is considered full compensation for completion of the work.

4. Method 4.

- a. For items of work for which specific unit prices are established by the contract, the volume of each type of rock riprap, including filter and bedding aggregate, is measured within the specified limits and computed to the nearest cubic yard by the method of average cross-sectional end areas.
- b. Payment is made at the contract unit price for each type of rock riprap, including filter and bedding. Such payment is considered full compensation for completion of the work.

5. Method 5.

- a. For items of work for which specific unit prices are established by the contract, the quantity of each type of rock riprap placed within the specified limits is computed to the nearest ton by actual weight. For each load of rock for riprap placed as specified, the contractor must furnish to the responsible engineer a statement-of-delivery ticket showing the weight to the nearest 0.1 ton.
- b. Payment is made at the contract unit price for each type of rock riprap, and includes compensation for any aggregate or geotextile installed as specified for filter or bedding. Such payment is considered full compensation for completion of the work.

6. Method 6.

- a. For items of work for which specific unit prices are established by the contract, the volume of each type of rock riprap is measured within the specified limits and computed to the nearest cubic yard by the method of average cross-sectional end areas.
- b. Payment is made at the contract unit price for each type of rock riprap and includes compensation for any aggregate or geotextile installed as specified for filter or bedding. Such payment is considered full compensation for completion of the work.

7. All methods.

- a. The following provision applies to all methods of measurement and payment. Compensation for any item of work described in the contract but not listed in the bid schedule is included in the payment for the item of work to which it is made subsidiary. Such items and the items to which they are made subsidiary are identified in section H.
- b. No separate payment is made for testing the gradation of the test pile. Compensation for testing is included in the appropriate bid item for riprap.

H. Items of work and construction details

Bid Item #7 – Class A IDOT certified RR-4 – Qty 54 tons

1. This bid item consist of delivery, excavation, placement and compaction of the riprap in designated rock trench checks as shown on the plans.
2. Contractor must provide COTR a IDOT certification for RR-4 class A riprap from quarry prior to deliver of any product.
3. Riprap shall be placed to ensure a workmanship like surface that is smooth and compacted with 2 passes of tracked placement equipment. Top of riprap surface shall be level with existing spillway channel requirements.
4. Placement will be located by inspector at station shown on plans.
5. Measurement and Payment – shall be by method 2 with the exception that the final quantities on material delivery tickets will indicate a minimum of designated tons were delivered. Final quantities installed will not be measured unless significant change to plans. It will be the contractor's responsibility to prove the quantity difference to COTR. Contractor will be responsible for removal of excess material. If there are no changes to the plans, then estimated volumes shall be used.